Initial entropy-production decrease for product-profile narrow beams in the Landau–Maxwell equation

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Abstract

We study the initial entropy-production derivative for the homogeneous Landau equation with Maxwell interaction in three velocity dimensions. The initial densities are finite mixtures of separated beams with transverse width $\sqrt{p}$, longitudinal width p, and independent local transverse and longitudinal profiles. For every fixed normalized configuration of positive polynomial–Gaussian profiles, we derive the full continuum expansion $D'_{0+}(f_p) = p^{-2}\sum_j w_j C_j + O(p^{-1})$. The coefficient contains longitudinal score skewness. A projection inequality and transverse moment–curvature bounds give $C_j \le -45/(14a_j^2)$ and $\sum_j w_j C_j \le -45/14$, where a_j is the local transverse variance. Thus the initial derivative is negative for all sufficiently small p, with a threshold depending on the fixed configuration. The proof differentiates the collision functional and controls mixture overlap, Gaussian tails and the initial solution derivative. It excludes this product-profile beam mechanism for a counterexample, while leaving universal and all-time monotonicity open.

Keywords: homogeneous Landau equation; Maxwell molecules; entropy production; narrow beams; anisotropic initial data.

1 Problem and relation to previous work

Entropy decay does not imply decay of entropy production. For the homogeneous Landau equation with Maxwell interaction, Tabary [5, Theorem 1.1] proves monotonicity when the directional temperatures are sufficiently well distributed, and hence after an explicit waiting time. In his time normalization, the three-dimensional sufficient condition is

$$T_{\max} - 1 \le c_3, \qquad c_3 = \frac{20 - 2\sqrt{70}}{15}. \tag{1.1}$$

Silvestre [4, Section 1.2] explicitly leaves the initial-time question for arbitrary Landau–Maxwell solutions open; his finite-angle Boltzmann counterexamples do not transfer to this operator. The counterexample of Junné, Winter and Yoldaş [3] concerns the Landau–Coulomb kernel instead.

We consider initial densities concentrated near finitely many points on a velocity axis. Their covariance is $\mathrm{diag}(p, p, 3-2p)$, so their largest directional temperature tends to three. They lie outside (1.1) for small p. Within each beam we allow a general fixed product profile, including nonradial transverse factors and asymmetric longitudinal scores. We derive an explicit leading coefficient for the actual initial entropy-production derivative and prove that it is strictly negative. The density family is an initial-data construction, not an invariant ansatz for the evolution.

The mechanism is a competition between transverse curvature, longitudinal curvature, and longitudinal score skewness. Projection of the longitudinal log-curvature onto the constant and score functions controls skewness without a symmetry assumption. Transverse moment identities then force enough curvature to make each redistributed local coefficient negative. A partner term cancels only after both beam-label averages; computing a component in isolation would miss this cancellation.

The elementary information inequalities, heat-flow identity and rotation–Gaussian representation used below are established tools; see Villani [6] and Delarue, Menozzi and Nualart [2, Proposition 2.3]. Our contribution is the specific product-beam collision coefficient, its sign bound, and its transfer to the actual continuum initial derivative. We make no claim of exhaustive priority over consequences implicit in earlier representations.

2 Equation, beam family and main result

2.1 Collision and time conventions

Velocity is $v \in \mathbb{R}^3$, with no spatial variable or external forcing. We use dimensionless velocity units with mass one, mean zero, and energy three. For a density f, write

$$\int f\,dv = 1, \qquad \int vf\,dv = 0, \qquad T = \int vv^{\mathsf{T}} f\,dv, \qquad \operatorname{Tr} T = 3. \tag{2.1}$$

The equation and kernel are

$$\partial_t f = Q(f) = \operatorname{div}_v \int a(v-w)\big[f(w)\nabla f(v) - f(v)\nabla f(w)\big]\,dw, \tag{2.2}$$

$$a(z) = \frac{|z|^2 I_3 - zz^{\mathsf{T}}}{2}. \tag{2.3}$$

All velocity integrals are over the full Euclidean space of the displayed dimension. For matrices, $A : B = \operatorname{Tr}(A^{\mathsf{T}}B)$. The entropy and production are

$$\begin{aligned} H(f) &= \int f \log f\,dv, \\ D(f) &= \frac{1}{2}\iint f(v)f(w)\,\Delta_f(v,w)^{\mathsf{T}} a(v-w)\Delta_f(v,w)\,dv\,dw, \\ \Delta_f(v,w) &= \nabla \log f(v) - \nabla \log f(w), \qquad \frac{d}{dt}H(f_t) = -D(f_t). \end{aligned} \tag{2.4}$$

Convolution of the quadratic kernel gives

$$A[f](v) = a(v) + \frac{3I_3 - T}{2}, \qquad Q(f) = \operatorname{div}(A[f]\nabla f) + \operatorname{div}(vf). \tag{2.5}$$

Since $\operatorname{div} A = -v$, the backward generator is $L_t\psi = A : \nabla^2\psi - 2v\cdot\nabla\psi$. It implies

$$T' = -6(T - I_3), \qquad T(t) = I_3 + e^{-6t}(T(0) - I_3). \tag{2.6}$$

These identities agree with Tabary [5, Lemmas 2.1–2.2]. For the kernel without the factor $1/2$, the rate is -12, as in Caja-Lopez et al. [1, Proposition 2]; thus $t_{\text{here}} = 2t_{\text{unscaled}}$.

2.2 Fixed local profiles

Fix a finite number of distinct real centers μ_j and positive weights w_j, with

$$\sum_j w_j = 1, \qquad \sum_j w_j\mu_j = 0, \qquad \sum_j w_j\mu_j^2 = 3. \tag{2.7}$$

For each label let P_j on $\mathbb{R}^2$ and R_j on $\mathbb{R}$ be centered probability densities with

$$\mathrm{Cov}(P_j) = a_j I_2, \qquad \mathrm{Var}(R_j) = \lambda_j > 0, \qquad a_j > 0, \qquad \sum_j w_j a_j = 1. \tag{2.8}$$

A positive polynomial–Gaussian density means a centered nondegenerate Gaussian multiplied by a polynomial globally bounded below by a positive constant, and divided by its integral. Centering and covariance conditions in (2.8) are imposed on the resulting density; its covariance need not equal that of its base Gaussian. Neither radiality of P_j nor parity of R_j is assumed.

Set $\bar{\lambda} = \sum_j w_j \lambda_j$ and define

$$\kappa_p = \left(1 - \frac{2p + \bar{\lambda} p^2}{3}\right)^{1/2}, \qquad m_j(p) = \kappa_p \mu_j, \tag{2.9}$$

$$F_{j,p}(v) = p^{-2} P_j(v_\perp/\sqrt{p}) R_j((v_3 - m_j(p))/p), \qquad f_p = \sum_j w_j F_{j,p}. \tag{2.10}$$

Here $v_\perp = (v_1, v_2)$ and p labels initial data, independently of evolution time t. We take sufficiently small $p > 0$ so that $\kappa_p \geq 1/2$. The Jacobian of the local coordinates is p^2. Equations (2.7)–(2.9) give exactly

$$\int v f_p \, dv = 0, \qquad T(f_p) = \mathrm{diag}(p, p, 3\kappa_p^2 + \bar{\lambda} p^2) = \mathrm{diag}(p, p, 3 - 2p). \tag{2.11}$$

For a single product factor $P(x)R(z)$, use the notation

$$g = \nabla \log P, \quad H_P = \nabla^2 \log P, \quad t_R = (\log R)', \quad h_R = (\log R)'',$$

$$J_P = \mathbb{E}_P \, \|H_P\|_{\mathrm{F}}^2, \quad U_4 = \mathbb{E}_P |x|^4, \quad K_2 = \mathbb{E}_P (x \cdot g)^2, \tag{2.12}$$

$$I_R = \mathbb{E}_R t_R^2, \quad J_R = \mathbb{E}_R h_R^2, \quad S_3 = \mathbb{E}_R t_R^3, \quad \mathcal{A} = 1 + 2a + U_4/4. \tag{2.13}$$

Subscripts j indicate the corresponding local factor.

Theorem 2.1 (Negative initial derivative for product beams)**.** *Assume* (2.7)–(2.8)*, with every local factor a positive polynomial–Gaussian density as specified above. The actual solution of* (2.2) *starting from f_p has an initial right entropy-production derivative, and*

$$D'_{0+}(f_p) = p^{-2} \sum_j w_j C_j + O(p^{-1}) \qquad (p \downarrow 0), \tag{2.14}$$

where, with $B_j = \mu_j^2 + 3$,

$$\begin{aligned} C_j = & -\frac{B_j^2}{2} J_{P,j} - 2\mathcal{A}_j J_{R,j} \\ & + (9 + 5\mu_j^2 - \mu_j^2 K_{2,j}) I_{R,j} - \mu_j a_j S_{3,j}. \end{aligned} \tag{2.15}$$

Every coefficient obeys

$$C_j \leq -\frac{5}{28} B_j^2 J_{P,j} \leq -\frac{45}{14 a_j^2}, \qquad \sum_j w_j C_j \leq -\frac{45}{14}. \tag{2.16}$$

Consequently $D'_{0+}(f_p) < 0$ for all sufficiently small $p > 0$. The remainder and the threshold depend on the complete fixed configuration. There is no assertion uniform over changing profiles, weights, widths, center gaps, or numbers of beams.

The inequality for the functional (2.15) also holds for regular positive product factors having the displayed finite moments and the integrations by parts used in Section 4. The stronger polynomial–Gaussian assumption ensures the actual-mixture and initial-solution assertions in (2.14).

3 Reconstruction from the collision functional

3.1 Exact first variation

Put $q = Q(f)/f$. At any time at which the differentiation is justified,

$$D'(f) = \iint f(v)f(w)\,[q(v)\Delta_f^{\mathsf T} a(v-w)\Delta_f + 2\nabla q(v)^{\mathsf T} a(v-w)\Delta_f]\,dv\,dw. \tag{3.1}$$

Indeed, the two density variations are equal after exchanging v, w, and the two score variations are also equal because $\partial_t \nabla \log f = \nabla q$. The prefactor $1/2$ in (2.4) then yields (3.1). From (2.5),

$$q = A[f] : (\nabla^2 \log f + \nabla \log f\, \nabla \log f^{\mathsf T}) + 3. \tag{3.2}$$

Thus (3.1) differentiates the original nonlinear collision functional, including its partner dependence; the covariance has not been held fixed along the evolution. Section 6 justifies this formula at the initial time for the densities in the theorem.

3.2 Leading local matrices and partner averages

For the leading calculation use the coordinates and derivative scaling

$$v = (\sqrt{p}\,x, m + pz), \qquad w = (\sqrt{p}\,y, n + p\zeta), \qquad \mathsf{J}_p = \mathrm{diag}(p^{-1/2}, p^{-1/2}, p^{-1}).$$

At $p = 0$, with $m = \mu_j$, $n = \mu_k$, the constant terms of $p\mathsf{J}_p A \mathsf{J}_p$ and $p\mathsf{J}_p a(v-w)\mathsf{J}_p$ are respectively

$$A_0(x; m) = \begin{pmatrix} BI_2/2 & -mx/2 \\ -mx^{\mathsf T}/2 & 1 + |x|^2/2 \end{pmatrix}, \qquad B = m^2 + 3, \tag{3.3}$$

$$K_0(x, y; m, n) = \begin{pmatrix} (m-n)^2 I_2/2 & -(m-n)(x-y)/2 \\ -(m-n)(x-y)^{\mathsf T}/2 & |x-y|^2/2 \end{pmatrix}. \tag{3.4}$$

Define the local quantities

$$L = \frac{\Delta P}{P} = \mathrm{Tr}\, H_P + |g|^2, \quad K = x \cdot g, \quad d_0 = 1 + |x|^2/2, \quad \ell = \frac{R''}{R} = h_R + t_R^2. \tag{3.5}$$

With $s = (g, t_R)$, the leading term of pq is

$$Q_0 = \frac{B}{2} L - mKt_R + d_0 \ell. \tag{3.6}$$

In this subsection $\mathbb{E}_*$ averages the partner profile and then all partner labels with their weights. Set $\bar{I} = \sum_k w_k I_{R,k}$. Centering, covariance and score integration by parts give

$$M_2 := \mathbb{E}_*[K_0(s - s_k)] = A_0 s + m e_3, \tag{3.7}$$

$$\begin{aligned} M_1 &:= \mathbb{E}_*[(s - s_k)^{\mathsf T} K_0 (s - s_k)] \\ &= \frac{B}{2}|g|^2 - mKt_R + 2mt_R + d_0 t_R^2 + \frac{|x|^2}{2}\bar{I} + c_m. \end{aligned} \tag{3.8}$$

The constant in the last line is

$$c_m = \frac{1}{2}\sum_k w_k [(m - \mu_k)^2 \mathbb{E}_{P_k}|g_k|^2 + 2a_k I_{R,k}].$$

It has no local-coordinate dependence. The score identities used here are $\mathbb{E}_{P_k} g_k = 0$, $\mathbb{E}_{P_k}[y_i (g_k)_r] = -\delta_{ir}$, and $\mathbb{E}_{R_k} t_{R,k} = 0$; local product independence is essential. Since $\mathrm{div}\, A_0 = -m e_3$, $\mathrm{div}\, M_2 + s \cdot M_2 = Q_0$. Integration of the gradient term in (3.1) therefore reduces the leading row to

$$\mathcal{R}_j = \mathbb{E}_{P_j R_j}[Q_0 M_1 - 2Q_0^2]. \tag{3.9}$$

Also $\mathbb{E} Q_0 = 0$, so c_m drops out.

3.3 Local integrations and the label cancellation

Suppress the label again. The longitudinal identities are

$$\begin{aligned}&\mathbb{E}t_R = 0, \quad \mathbb{E}h_R = -I_R, \quad \mathbb{E}(h_R t_R) = -S_3/2, \quad \mathbb{E}(h_R t_R^2) = -\mathbb{E}t_R^4/3,\\ &\mathbb{E}\ell = 0, \quad \mathbb{E}(t_R\ell) = S_3/2, \quad \mathbb{E}(t_R^2\ell) = 2\mathbb{E}t_R^4/3, \quad \mathbb{E}\ell^2 = J_R + \mathbb{E}t_R^4/3.\end{aligned} \tag{3.10}$$

For example, integrate the derivatives of Rt_R, Rt_R^2 and Rt_R^3. In the transverse variables,

$$\begin{aligned}&\mathbb{E}K = -2, \quad \mathbb{E}(Kd_0) = -2 - 4a, \quad \mathbb{E}d_0 = 1 + a, \quad \mathbb{E}d_0^2 = \mathcal{A},\\ &\mathbb{E}L = 0, \quad \mathbb{E}(Ld_0) = 2, \quad \mathbb{E}(L|x|^2) = 4,\\ &\mathbb{E}(L|g|^2) - 2\mathbb{E}L^2 = -2J_P.\end{aligned} \tag{3.11}$$

The last identity is the usual heat-flow Fisher dissipation identity: for $\partial_\tau P = \Delta P$, differentiating $\mathbb{E}|g|^2$ gives $\mathbb{E}(L|g|^2) + 2\mathbb{E}(g \cdot \nabla L)$, with $\mathbb{E}(g \cdot \nabla L) = -\mathbb{E}L^2$; integration by parts also gives the derivative as $-2J_P$. The polynomial–Gaussian factors justify both calculations directly, without log-concavity.

Substitution of (3.10)–(3.11) into (3.9) cancels all fourth-score moments and yields

$$\mathcal{R}_j = -\frac{B_j^2}{2}J_{P,j} - 2\mathcal{A}_j J_{R,j} + (3 + 5\mu_j^2 - \mu_j^2 K_{2,j})I_{R,j} - \mu_j a_j S_{3,j} + B_j\bar{I}. \tag{3.12}$$

The difference between this row and C_j in (2.15) is $B_j\bar{I} - 6I_{R,j}$. It vanishes after the final label sum:

$$\sum_j w_j(B_j\bar{I} - 6I_{R,j}) = 6\bar{I} - 6\bar{I} = 0, \tag{3.13}$$

because $\sum_j w_j B_j = 6$. This establishes the asserted leading coefficient before passing from the separated component calculation to the actual mixture.

4 A negative bound for every local coefficient

Lemma 4.1 (Longitudinal projection)**.** *For a regular positive probability density R with the moments and integrations by parts above,*

$$J_R \geq I_R^2 + \frac{S_3^2}{4I_R}. \tag{4.1}$$

Proof. An integrable positive density has $I_R > 0$. In $L^2(R\,dz)$, the functions 1 and t_R are orthogonal, with squared norms 1 and I_R. Their inner products with h_R are $-I_R$ and $-S_3/2$ by (3.10). The squared norm of the orthogonal projection of h_R gives (4.1). □

For any $\mathcal{A} > 0$, completing the skewness square now gives

$$\begin{aligned}-2\mathcal{A}J_R - maS_3 &\leq -2\mathcal{A}I_R^2 - \frac{\mathcal{A}}{2I_R}\left(S_3 + \frac{maI_R}{\mathcal{A}}\right)^2 + \frac{m^2a^2}{2\mathcal{A}}I_R\\ &\leq -2\mathcal{A}I_R^2 + \frac{m^2a^2}{2\mathcal{A}}I_R.\end{aligned} \tag{4.2}$$

This estimate retains both signs and arbitrary magnitudes of score skewness; no symmetry of R has been introduced.

Lemma 4.2 (Transverse curvature and fourth moment)**.** *Let P be a regular positive centered density on $\mathbb{R}^2$ with covariance aI_2 and the stated finite moments and integrations by parts. Then $u = U_4/a^2 > 4$, and*

$$J_P \geq \frac{2}{a^2}, \qquad K_2 \geq 4 + \frac{16}{u-4}, \qquad K_2 \leq 4 + \sqrt{2J_P U_4}. \tag{4.3}$$

In particular,

$$J_P U_4 \geq 14, \qquad J_P \mathcal{A} \geq 7/2. \tag{4.4}$$

Proof. Since $\mathbb{E}|x|^2 = 2a$ and $|x|^2$ is not constant almost surely under a positive density, $U_4 > 4a^2$. Integration by parts gives $\mathbb{E}(x \cdot g) = -2$; hence Cauchy–Schwarz implies $\mathbb{E}|g|^2 \geq 2/a$. Also $\mathbb{E}H_P = -\mathbb{E}(gg^{\mathsf{T}})$. Jensen and the matrix trace inequality give

$$J_P \geq \|\mathbb{E}H_P\|_{\mathrm{F}}^2 \geq \tfrac{1}{2}(\mathbb{E}|g|^2)^2 \geq 2/a^2.$$

Writing $r^2 = |x|^2$, another integration by parts gives $\mathbb{E}(Kr^2) = -8a$. Thus

$$\mathrm{Cov}(K, r^2) = -4a, \quad \mathrm{Var}(K) = K_2 - 4, \quad \mathrm{Var}(r^2) = U_4 - 4a^2.$$

Their covariance inequality proves the lower bound on K_2. Pointwise $K^2 \leq r^2|g|^2$, while

$$\mathbb{E}(r^2|g|^2) = 4 - \mathbb{E}(r^2 \operatorname{Tr} H_P) \leq 4 + \sqrt{2J_P U_4}.$$

This proves the upper bound. Combining them gives

$$J_P U_4 \geq \max\left\{2u, \frac{128}{(u-4)^2}\right\} \geq 14.$$

For $u \geq 7$ use $2u \geq 14$; for $4 < u \leq 7$ use $128/(u-4)^2 \geq 128/9 > 14$. Finally $\mathcal{A} \geq U_4/4$ proves the last assertion. □

Proof of the coefficient bound in Theorem 2.1. Fix one label and write $m = \mu_j$, $B = m^2 + 3$. Since $a^2/(2\mathcal{A}) \leq 2/u$, Lemma 4.2 implies

$$5 - K_2 + \frac{a^2}{2\mathcal{A}} \leq 1 - \frac{16}{u-4} + \frac{2}{u} < 1. \tag{4.5}$$

Equations (2.15) and (4.2) consequently give

$$C_j \leq -\tfrac{1}{2}B^2 J_P - 2\mathcal{A} I_R^2 + cI_R, \qquad c \leq 9 + m^2 \leq 3B.$$

As $I_R \geq 0$, maximization of the last quadratic and $J_P\mathcal{A} \geq 7/2$ yield

$$\begin{aligned} C_j &\leq -\frac{1}{2}B^2 J_P + \frac{9B^2}{8\mathcal{A}} \leq -\frac{5}{28}B^2 J_P \\ &\leq -\frac{5}{14}\frac{B^2}{a^2} \leq -\frac{45}{14a^2}. \end{aligned} \tag{4.6}$$

Jensen's inequality for $a \mapsto a^{-2}$, together with $\sum_j w_j a_j = 1$, gives $\sum_j w_j a_j^{-2} \geq 1$ and completes (2.16). The numerical constant is a convenient nonsharp bound; no optimality is asserted. □

5 Passage from component coefficients to the mixture

5.1 Polynomial scaling of the component calculation

In a component-pair integral use the true whole-mixture diffusion matrix $A = a(v) + (3I_3 - T(f_p))/2$, but replace the logarithmic derivatives at v, w by those of $F_{j,p}, F_{k,p}$, respectively. Let $\mathcal{P}_{jk}(p)$ denote this replacement in (3.1), including the corresponding replacements in q and ∇q.

For temporarily fixed centers m, n, both full scaled matrices $p\mathsf{J}_p A \mathsf{J}_p$ and $p\mathsf{J}_p a(v-w)\mathsf{J}_p$ are polynomials in p of degree at most two. The local scores and their derivatives are independent of p. Formula (3.2) shows that pq is also a polynomial of degree at most two, including the term $3p$. In the gradient term, $\nabla_v = \mathsf{J}_p \nabla_{x,z}$; hence p^2 times the full replaced integrand has degree at most four. Every coefficient is integrable under the fixed polynomial–Gaussian product factors. Replacing m, n by $\kappa_p \mu_j, \kappa_p \mu_k$, with $\kappa_p = 1 + O(p)$, changes the leading coefficient by $O(p)$ and leaves the other coefficient integrals bounded. Therefore

$$\sum_{j,k} w_j w_k \mathcal{P}_{jk}(p) = p^{-2} \sum_j w_j C_j + O(p^{-1}). \tag{5.1}$$

The normalization identities for the limiting centers are used at $p = 0$, not imposed on the finite-p centers separately from the internal variances.

5.2 Posterior overlap and full-space tails

We give the bounds that permit the score replacements just made. Let $d_* = \min_{j \neq k} |\mu_j - \mu_k| > 0$. Let $\sigma_{\min}, \sigma_{\max}$ be the smallest and largest base Gaussian standard deviations of the longitudinal factors, and let $b_{\max}$ bound all transverse base Gaussian covariance eigenvalues. All these are fixed positive constants. Choose $r_0 = d_* \sigma_{\min}/(16\sigma_{\max})$, and use the physical core

$$\Omega_j(p) = \{\, |v_3 - m_j(p)| \le r_0, \quad |v_\perp| \le 1 \,\}. \tag{5.2}$$

For sufficiently small p, every other center is separated from this core in the longitudinal coordinate. Comparison of the Gaussian exponents, allowing for unequal base variances, gives on $\Omega_j(p)$

$$\frac{\text{base Gaussian of } F_{k,p}(v)}{\text{base Gaussian of } F_{j,p}(v)} \le C \exp(C/p - c/p^2), \qquad k \neq j.$$

The fixed choice of r_0 makes the coefficient of p^{-2} strictly negative. The transverse ratio contributes at most $\exp(C/p)$. Positive lower bounds on the defining polynomials allow their ratio to contribute only a polynomial in p^{-1} and $|v|$. Thus the wrong-component posterior weights are exponentially small on each core. Raw derivative ratios through order three, and hence the logarithmic derivatives entering $q, \nabla q$, have polynomial bounds in p^{-1} and $|v|$. Differentiation of posterior weights preserves the exponential estimate, with an additional polynomial factor.

Under component j, the complement of (5.2) has a polynomial-weighted Gaussian tail bounded by a power of p^{-1} times

$$C \exp\left(-\frac{r_0^2}{4\sigma_{\max}^2 p^2}\right) + C \exp\left(-\frac{1}{4 b_{\max} p}\right). \tag{5.3}$$

One half of each Gaussian exponent supplies the displayed tail and the other supplies any required polynomial moment. For the mixture, $\partial^\alpha f_p / f_p$ is a posterior-weighted average of the component raw derivative ratios. Consequently the exact and replaced pair integrands have global polynomial bounds of the same type, without requiring a uniform positive lower

bound on f_p over velocity space. Split each component-pair integral into its two cores and their complement. For fixed configuration-dependent $C, K, c > 0$, this gives

$$\left| \mathcal{V}(f_p) - \sum_{j,k} w_j w_k \mathcal{P}_{jk}(p) \right| \le C p^{-K} e^{-c/p} = o(p^{-1}), \tag{5.4}$$

where $\mathcal{V}(f)$ denotes the right-hand side of (3.1). This is an analytic full-space estimate, not a velocity-grid or finite-difference approximation.

6 The actual initial right derivative

It remains to identify $\mathcal{V}(f_p)$ with the actual solution derivative at each fixed admissible p. All constants in this section may depend on p; no uniform ellipticity as $p \downarrow 0$ is used. Put

$$C(t) = \frac{3I_3 - T(t)}{2}, \qquad B_{ir} = \frac{v_i \partial_r - v_r \partial_i}{\sqrt{2}} \quad (i < r).$$

For the covariance (2.11), $C(t)$ is positive definite near zero. The backward generator in (2.5) satisfies

$$L_t = A : \nabla^2 - 2v \cdot \nabla = \sum_{i<r} B_{ir}^2 + C(t) : \nabla^2 - v \cdot \nabla. \tag{6.1}$$

The angular diffusion has an orthogonal Brownian flow O_t, normalized to generate $\sum B_{ir}^2$. Variation of constants gives a process

$$V_t = e^{-t} O_t V_0 + G_t, \tag{6.2}$$

where, conditional on the angular path, G_t is a centered Gaussian independent of V_0, with covariance

$$\mathrm{Cov}(G_t \mid O) = 2 \int_0^t e^{-2(t-s)} O_t O_s^{\mathsf{T}} C(s) O_s O_t^{\mathsf{T}} \, ds. \tag{6.3}$$

This is the established rotation–Gaussian mechanism of Delarue, Menozzi and Nualart [2, Proposition 2.3], here reconstructed with the kernel and time convention (2.3).

For completeness this prescribed-covariance process is self-consistent. Its second moment $M(t)$ obeys

$$M' = (\operatorname{Tr} M + 3) I_3 - 5M - T(t), \qquad (\operatorname{Tr} M)' = 6 - 2 \operatorname{Tr} M.$$

Thus $\operatorname{Tr} M = 3$, and $(M - T)' = -5(M - T)$ with zero initial value gives $M = T$. Its density therefore solves the nonlinear equation (2.2). Conversely, a regular normalized solution has the covariance (2.6) and hence this same linear generator; the affine stochastic equation has a unique law.

The density formula is

$$f_t(v) = e^{3t} \mathbb{E}[f_p(e^t O_t^{\mathsf{T}}(v - G_t))]. \tag{6.4}$$

For fixed p, every raw derivative ratio $\partial^\alpha f_p / f_p$, $|\alpha| \le 4$, grows at most polynomially, and the initial density has moments of every order. Differentiate (6.4). The derivative of the affine argument is a bounded linear map for small time. Jensen's inequality with the posterior density weights, followed by change of variables, therefore bounds, uniformly on a sufficiently small time interval,

$$\int f_t(v)(1 + |v|)^k \left| \frac{\partial^\alpha f_t(v)}{f_t(v)} \right|^s dv, \qquad |\alpha| \le 4, \quad k \ge 0, \quad s \ge 1. \tag{6.5}$$

The bounds reduce to initial polynomial-weighted derivative moments and bounded moments of the conditional Gaussian noise. Orthogonality controls the angular factor. Extra velocity powers in (6.5) give uniform integrability at infinity.

As $t \downarrow 0$, $O_t \to I_3$ and $G_t \to 0$, so density derivatives converge locally to the initial derivatives. Positivity gives local convergence of their ratios. Compact/tail splitting and Hölder's inequality, using (6.5), now give convergence of both $D(f_t)$ and $\mathcal{V}(f_t)$. At positive times, cutoff differentiation and integration by parts prove $dD(f_t)/dt = \mathcal{V}(f_t)$; the same bounds remove the cutoffs. Integration from ε to t and passage $\varepsilon \downarrow 0$ yield

$$D(f_t) - D(f_p) = \int_0^t \mathcal{V}(f_s)\, ds, \qquad D'_{0+}(f_p) = \mathcal{V}(f_p).$$

Together with (5.1)–(5.4), this proves (2.14). The bound (2.16) then completes Theorem 2.1.

7 Interpretation and limitations

Separated counterstreaming beams with strong temperature anisotropy are a natural test of initial entropy-production monotonicity. The theorem shows that arbitrary fixed non-Gaussian product profiles do not produce a positive initial derivative in the scaling (2.10), even when longitudinal score skewness is present. For this beam geometry, a positive sequence as $p \downarrow 0$ would have to leave at least one of the fixed-configuration or local-independence hypotheses. These are possible escape routes, not assertions that a counterexample exists there.

The conclusion concerns only the initial right derivative. It does not establish monotonicity at later times, an invariant mixture family, arbitrary correlated local profiles, rough initial data, or a statement uniform over varying configurations. The remainder constant and finite threshold are not computed explicitly, and 45/14 is not sharp. The functional sign estimate holds under broader integration-by-parts conditions than the actual-mixture theorem; we do not extend the latter without its tail and regularity assumptions. No Boltzmann or Coulomb conclusion follows by changing the kernel in the present proof. The universal Landau–Maxwell initial-time monotonicity problem remains unresolved by this result.

Reproducibility

The accompanying original Python/SymPy programs reconstruct the leading two-particle coefficient from explicit product-profile moment identities, check the partner redistribution and square completion, and verify the degree bounds of the full scaled matrices. The supplement specifies the environment and one command for replay. Exact algebra checks support the displayed calculation; the integrations by parts, mixture estimates and solution differentiability are justified in the text. No numerical scan is used as a proof of the sign, and no experimental data are used.

Use of generative AI tools

OpenAI Codex was used in developing the analytic argument, generating symbolic checking code, performing literature searches and internal checks, and drafting this article. The disclosure applies throughout the argument and to the accompanying code. AI checks are not external peer review and can share errors. Human scientific review, confirmation of authorship and affiliation, and acceptance of responsibility for the complete manuscript remain pending for this local draft.

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